Assumptions behind eqs. (1)–(20), audited on the campaign data

eq.	assumption	status	what was measured	moves $\hat{\theta}$?	moves σ?
§2, §3	b deterministic; noise perturbs $\mathbf{H}$	HARMLESS	speciality pins $\mathbf{H}=\mathbf{I}$ to 10^{-15}, so $\delta\mathbf{H}=0$ and ALL randomness is in b	none	none
eq. (2)	noise spatially uncorrelated, $\delta_{pq}\Gamma$	VIOLATED	$\rho=0.92$ / 0.94 at one grid step; $\kappa=10.5$ / 18.5 correlated points	+2% on D_{22}	×3.8
eq. (2)	Γ homogeneous over the ROI	PARTIAL	correlation is anisotropic: $\rho=0.73$ along x vs 0.89 along y	not isolated	in κ
§8	$\mathbf{MY}\approx 0$: basis captures field	PARTIAL	σ tracks the load ($r=0.88$–0.99), so Γ is EFFECTIVE error, not DIC noise	$\leq 1\%$	×1
§14	first-order plug-in; $\mathbf{E}^*(\theta)$ from data	PARTIAL	bootstrap under eq. (2)'s own noise model vs eq. (20), $\kappa=1$: ratio 1.1–3.5	none detected	×2.1
§19	eq. (20) treats $\hat{\Gamma}$, $\mathbf{C}$ as fixed	PARTIAL	the derivation says so itself; jackknife/eq.(20) $=25.6$, 8.6, 8.2, 2.8	none	×2–6
§1	rank$(\mathbf{H}_{t,t})=p$ per load step	VACUOUS	$\mathbf{H}=\mathbf{I}$ BY CONSTRUCTION, so the rank test always passes and detects nothing	hides D_{12}, D_{66} blindness	none
eq. (12)	experiments independent	VACUOUS	untestable with two coupons, no repeats; a shared calibration error would not show	unknown	unknown
§1	ONE common θ fits all experiments	VIOLATED	the coupons disagree on D_{66} by 26.8σ, on D_{22} by 382σ of their own error bars	dominant for D_{66}	not represented
implicit	no model error: plate model exact	DOMINANT	$\chi^2/\mathrm{dof}=830$: the noise model explains 0.1% of the residual variance	this IS the D_{66} error	×28.8 min.

Read the last two columns. Only two rows move the ANSWER, and the one that moves it most — the bottom row — is the single assumption the derivation never writes down. Everything above it changes the error bar but not the number.